

The Manufacturing of Artifacts

Low Barrier to Entry Methodologies in Digital Manufacturing

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Abstract

The proliferation of consumer-based mobile phone LiDAR scanners/cameras and DYI 3D printers has lowered the entry barriers towards integrating digital workflows into a foundations studio art curriculum. From citizen scientists scanning coastlines with LiDAR-based mobile phone apps to the integration of 3D-printing into primary and secondary school STEM pedagogies, today's university students are more accustomed to and accepting of, these technologies as part of everyday life. Can this acceptance be leveraged toward the integration of digital thinking into studio art curriculums that often eschew technology due to perceived costs and learning curves?

This abstract demonstrates an exercise and results that exploit a low barrier to entry methodology by introducing photogrammetry, mesh editing, and 3D printing into a studio arts-based curriculum, making visual the influence and impact of each process on the final construct. During a two-week workshop for bachelor-level university students, experiments in scanning, photogrammetry, and 3D modeling/editing are integrated into a design pipeline exploring ceramic slip-casting. Using mobile phones, readily available scanning/photogrammetry software, basic modeling software, and off-the-shelf 3D printers, students first capture an archeological fragment (photogrammetry), next format the fragment for 3D printing (digital model manipulation and editing), 3D print the fragment (contouring), before casting a negative mold from plaster before making the final positive cast through slip-casting. The resulting products bear the moniker "artifact" as each product "wears" the influence and characteristic of every process in the design pipeline, including the student's understanding and mastery of each process.

Keywords

Photogrammetry, 3D Printing, Studio arts, Educational workshop

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1 INTRODUCTION

The practices of design and architecture are based in workflows, from initial conceptual designs all the way through to manufacturing and final construction. Each process embedded within the sequence of a workflow contains input parameters and output qualities, characteristics that leave their "thumbprint" on the final piece of work. And while the term workflow appears to describe a linear process, workflows are often anything but, requiring feedback loops and other workflows operating in tandem to articulate a final product. Understanding these characteristics is critical towards harnessing the power of a workflow and an important part of architectural thinking and education.

In this digital workshop workflows are tested beginning with virtual information that is continually refined and shaped by every process within the sequence. We start with a scanned fragment from a physical piece, formatting it, 3D-printing it, and making molds from it before casting back into those molds. And because workflows are not linear, throughout the process an analog support system will be continually designed and redesigned to display the final artifact through precise joinery. The resulting artifact and joint will broadcast all of the characteristics of the workflow, making evident our "thumbprints" on this process.

1.1 CAPTURING INITIAL FRAGMENTS

Day 01: Using a digital camera (handheld or phone-based) capture an interesting fragment from a minimum of three different pieces exhibited at the museum. For clarity, a fragment means a portion of an exhibited piece – not the entire piece. Photogrammetry is still an emergent and continually evolving tool. As such, each fragment piece captured should be exhibited outside of a glass or acrylic case as this will limit reflection/glare and allow easier access to the piece. Remember that more photographs = a better capture. . . how many photographs needed for a proper capture must be continually tested. Finally, captures should be processed through the photogrammetry software while at the museum in order to determine if a capture must be redone

1.2 VISUALIZING FRAGMENTS

Day 02: Use the morning to determine the best view for each of your three (minimum) captures. Take screen shots (either in the photogrammetry software or Blender) of these views, saving an individual jpg for each of the following:

- A point cloud representation of each capture
- A wireframe mesh of each capture
- A rendered mesh of each capture

Have these jpgs ready on a thumb drive for examination in the afternoon.

1.3 FORMATTING FRAGMENTS

(Days 03 and 04) After selecting ONE of your many captures, format the mesh for 3D printing via Blender. You are welcome to continue to work with the original mesh, or convert/evolve the mesh into a series of surfaces for 3D printing, meaning making the surfaces “watertight” and legible to the CURA 3D printing software. The captured fragment must fit within one of three bounding box sizes: 25mm x 25mm x 50mm, 50mm x 50mm x 50mm, or 75mm x 75mm x 50mm.

Why? These dimensions represent the sizes of the preprinted molds built for you. You are welcome to shrink the scale of your captured fragment to fit within one of these dimensions, or subdivide the captured fragment into a series of modules that fit within these bounding dimensions. Just remember that the more modules created, the more printing time needed. Finally, with digital geometries formatted for printing, send all captured fragments to the 3D printers.

1.4 JOINING FRAGMENTS TO ARTIFACTS

(Days 05 and 06) The 3D prints and plaster fragment molds will take time to harden, leaving you time to develop and design the second part of this exercise, a wooden structure that joins to the fragment. You are free to use scale timber (sticks and staves) or carve out of larger pieces of timber. Critical in this part of the exercise are the following conditions:

- The wooden system must join to the (eventual) slip cast fragment in a purposeful way, meaning the joint looks intentional
- The wooden system should not be seen as only a “display” but more importantly as an integral part of the overall design construction

The design, testing, and finalizing of this wooden joint will take place at the same time as the plaster molds are poured and the final fragment is slip cast, meaning some degree of flexibility in the wooden design is needed in order to contend with tolerances and inaccuracies of the manufacturing process.

1.5 CASTING FRAGMENT MOLDS

(Days 05, 06, and 07) The morning should be used to clean up and remove and support structures or other 3D printed artifacts not meant to be used in the plaster molds. After lunch each mold is setup and plaster is poured, letting it harden over the weekend before releasing the molds on Monday.

1.6 SLIP CASTING ARTIFACT REPRODUCTIONS

Once the plaster molds have sufficiently dried, the slip casting material is mixed and poured into the plaster molds – producing an evolved reproduction of the scanned fragment. This process should be repeated several times over, testing different wall thicknesses in slip by varying the time left to cure in the plaster molds.

1.7 INITIAL REFLECTION ON EXERCISE

This exercise was undertaken by undergraduate architecture students with little to no experience in photogrammetry, mesh manipulation, 3D printing, nor slip casting. By integrating common everyday tools (mobile phones for photogrammetry) and the fundamental of mesh editing (Blender) into a physical manufacturing process, students appeared to better grasp both the technologies and their impact on a tangible product. The extended paper and presentation will include both recorded and anecdotal comments from students about their experiences learning these new technologies and how these introductions lead to further computational explorations in recording, representation, and digital manufacturing.